7 Petala Langit · Memory Architecture

> *A complete reference for the seven-layer memory substrate of arifOS — from ephemeral intent capture (L0) to cryptographically-sealed, append-only decisions (L6). For Fahmi (technical onboarding) and for general readers (sections marked 📖).*

Contents

0. Plain-English Summary

The arifOS memory system has **7 layers**, ranging from the lightest (a fresh chat message) to the heaviest (a permanently-sealed, cryptographically-signed decision). Think of it like a tree — leaves at the top (fall off daily), roots at the bottom (permanent, unshakeable).

Why 7? Because each type of "memory" has different needs:

- New chat messages should be **fast to forget** (no need to store forever)
- Important history should be **permanent and uneditable** (audit trail)
- Some memory is **volatile** (Redis, lost on restart), some is **persistent** (Postgres, ACID guarantees)

Imagine 7 types of books in a library:

1. **Post-it notes** — fresh chat, throw away tomorrow
2. **Memos** — working notes, kept for a week
3. **Card catalogue** — search by meaning
4. **Filing cabinet** — organized records
5. **Knowledge map** — how things connect
6. **Diary** — how state changed over time
7. **Notary seal** — legally witnessed, cannot lie

KEY INSIGHT

OpenCode (the code tool, now available in Telegram via `/forge`) only uses post-it notes + filing cabinet (ephemeral + project files). It **cannot access the notary seal** (VAULT999). This is not a restriction — it's a safety design, so the code tool can never "create false history."

1. Architecture Diagram — The Machine View

```
              ┌──────────────────────────────────────────────────────────┐
              │                SOVEREIGN INTENT LAYER (L0)                │
              │    Arif's typed word — ephemeral, highest authority       │
              │         "Apa yang aku nak" (what I want)                  │
              └──────────────────────────────────────────────────────────┘
                                         │
                                         │
                                         │ [envelope: intent, scope,
                                         │  reversibility, risk_band]
                                         ▼
              ┌──────────────────────────────────────────────────────────┐
              │   CONSTITUTIONAL FLOORS (F1→F13) — sit between L0/L1     │
              │   AMANAH → TRUTH → SEAL → SCOPE → ... → SOVEREIGN         │
              │   888_HOLD gates irreversible actions                    │
              └──────────────────────────────────────────────────────────┘
                                         │ [all floors pass]
                                         ▼
  ┌────────────────────────────────────────────────────────────────────────┐
  │  L1 · REDIS (ephemeral cache)                                          │
  │      task queue, session context, working state, dedup, rate limits   │
  │      TTL: 1h–24h · lossy by design · volatile                           │
  └────────────────────────────────────────────────────────────────────────┘
                              │
                              │
                 [12h curator pass — promote persistent items]
                              │
                              ▼
  ┌────────────────────────────────────────────────────────────────────────┐
  │  L3 · SUPABASE (PostgreSQL — durable, ACID, canonical)                  │
  │      sessions · messages · tasks · decisions · vault_seals · agents · envelopes │
  │      RLS-enforced · source of truth for relational data                │
  └────────────────────────────────────────────────────────────────────────┘
      │                                   │                              │
      │                                   │                              │
      │ [re-embed nightly]                │ [event-driven sync]          │ [12h compute]
      ▼                                   ▼                              ▼
  ┌──────────────────┐        ┌──────────────────────┐       ┌─────────────────────┐
  │ L2 · QDRANT      │        │ L4 · GRAPHITI         │       │ L5 · HYSTERESIS      │
  │ Vector embeddings│        │ Knowledge graph       │       │ Path-dependent state │
  │ 1536-dim cosine  │        │ Entity-relationship   │       │ GROWTH/PLATEAU/      │
  │ Semantic search  │        │ Multi-hop traversal   │       │ REVERSION/COLLAPSE   │
  └──────────────────┘        └──────────────────────┘       └─────────────────────┘
                                                                         │
                                                                         │
                                                             [only SEAL-worthy
                                                              events flow down]
                                                                         │
                                                                         ▼
                                                          ┌────────────────────────┐
                                                          │ L6 · VAULT999 (SEALED)  │
                                                          │ SHA3-512 chained        │
                                                          │ F1-F13 constitutionally │
                                                          │  gated · append-only    │
                                                          │ No code path writes back│
                                                          │ Final, witnessed, forever│
                                                          └────────────────────────┘
```

CRITICAL INVARIANT

Data flows **upward** (L0 → L6), gaining permanence at each step. Once sealed to **VAULT999**, there is **no code path that writes back downward** — by architecture, not by policy.

2. Layer-by-Layer Spec

L0 · NIAT (Sovereign Intent)

Capture Arif's typed intent before any processing.

PROPERTY	VALUE
Storage	RAM only (within Hermes session)
Persistence	None — gone when session ends
Authority	Highest human authority (Arif = SOVEREIGN)
Format	Natural language + structured envelope

Design note: L0 is *not* persisted. If the session dies, intent dies. Intent is volatile; decisions are not. Decisions get sealed (L6); intent doesn't need to be, because intent is the input, not the verdict.

L1 · REDIS (Ephemeral Cache)

Fast in-memory state for active tasks, sessions, agent coordination.

PROPERTY	VALUE
Storage	Redis 7 (in-memory, optional AOF)
Persistence	Configurable (default: none — pure ephemeral)
TTL	1h–24h depending on data type
Use cases	Task queue, session cache, rate limits, dedup

Design note: L1 is intentionally lossy. If Redis crashes, the *worst* that happens is active tasks need to restart. Decisions never live here.

L2 · QDRANT (Vector Embeddings)

Similarity search — given a query, find semantically related past content.

PROPERTY	VALUE
Storage	Qdrant (Rust-based vector DB)
Embedding model	`text-embedding-3-small` (1536-dim) or `bge-large-en` (1024-dim)
Distance metric	Cosine similarity
Indexing	HNSW for sub-millisecond search

Design note: L2 is *derived* — embeddings are computed from text that exists in L3 or L6. The vector is a secondary index, not primary truth. If Qdrant loses data, it can be rebuilt from L3/L6.

L3 · SUPABASE (Relational + ACID)

Canonical, queryable, ACID-compliant record store.

PROPERTY	VALUE
Storage	Supabase (managed PostgreSQL 15)
Persistence	Durable, backed up daily
Authority	Source of truth for relational data
Access	RLS-enforced (Row Level Security)

Major tables: `sessions` , `messages` , `tasks` , `decisions` , `vault_seals` , `agents` , `envelopes` , plus organ-specific (`geox_claims` , `wealth_transactions` , `well_vitals`).

Design note: L3 is the **canonical** layer. If there's a dispute about "what is the record?", L3 wins. Vector search (L2) and graph traversal (L4) are *views* on top of L3. Sealed VAULT999 (L6) is anchored to L3 but cryptographically independent.

L4 · GRAPHITI (Knowledge Graph)

Multi-hop entity-relationship queries that SQL can't do well.

PROPERTY	VALUE
Storage	Graphiti (custom, on Neo4j/Memgraph)
Schema	Nodes (entities) + Edges (relationships) + Episodes (temporal)
Use case	"How does GEOX connect to arifOS through the constitutional layer?"

Edge types: `GOVERNS`, `IMPLEMENTS`, `PRODUCED`, `REFERENCED`, `CHALLENGED`, `WITNESSED`, `OWNS`, `SIGNED`.

Design note: L4 is **derived** from L3. Every node/edge has a primary key in L3; L4 is a graph view for traversal. If graph data is lost, it can be rebuilt from L3 + L6.

L5 · HYSTERESIS (Path-Dependent State)

Track state evolution — how the system's state has shifted over time.

PROPERTY	VALUE
Storage	Supabase `hysteresis_ledger` + LRU cache
Schema	State machine: GROWTH / PLATEAU / REVERSION / COLLAPSE
Use case	"Are we growing, plateauing, or collapsing in capability?"

State transitions:

- `Tool success rate > 80% for 7 days` → PLATEAU → GROWTH
- `Constitutional floor violation (F1–F13)` → * → COLLAPSE
- `Repeated retries without progress` → GROWTH → REVERSION
- `Capacity headroom > 50% for 30 days` → PLATEAU → GROWTH
- `Audit chain broken` → * → CRITICAL

Design note: L5 is a **meta-layer** — it observes the system and characterizes its trajectory. It does not store domain data; it stores **system state evolution**.

L6 · VAULT999 (Sealed, Immutable)

Final, cryptographically-provable, constitutionally-gated record of decisions.

PROPERTY	VALUE
Storage	Append-only JSONL files + Supabase mirror
Hash chain	SHA3-512, each entry references previous hash
Signatures	F1–F13 constitutional gates, witness consensus
Access	Read = auth users; **Write =** `vault_seal` **+** `ack_irreversible=True`
Retention	Forever (immutable)

Seal schema:

```
{
  "seal_id": "vault-2026-06-07-001337",
  "envelope_id": "uuid-v4",
  "actor_id": "arif",
  "actor_signature": "sig-...",
  "nonce": "unique-nonce-per-seal",
  "constitutional_chain_id": "F1→F2→...→F13",
  "judge_state_hash": "sha256-of-judge-verdict",
  "witness_type": "human | ai | council",
  "payload": "{...decision content...}",
  "prev_hash": "sha3-512-of-previous-seal",
  "seal_hash": "sha3-512-of-(payload + prev_hash + nonce)",
  "sealed_at": "2026-06-07T04:33:00+08:00"
}
```

Design note: L6 is **the ground truth of decisions.** If L3 says one thing and L6 says another, **L6 wins.** L6 is independent of L3 by design — even if Supabase is compromised, the SHA3 chain can be verified.

3. Truth Flow

```
  HUMAN (Arif)
       │
       │ [typed intent + envelope]
       ▼
 ┌──────────┐
 │   L0     │  ephemeral, volatile
 │  NIAT    │  "what I want"
 └────┬─────┘
      │
      │ [constitutional floors F1-F13 gate]
      ▼
 ┌──────────┐
 │   L1     │  Redis cache, working state
 │  REDIS   │  TTL: 1h–24h
 └────┬─────┘
      │
      │ [12h curator pass — promote persistent]
      ▼
 ┌───────────────────────────────────┐
 │      L3 · SUPABASE                │  ACID, durable
 │  canonical record store           │  RLS-enforced
 └──┬────────────────┬───────────────┘
    │                │
    │ re-embed       │ sync
    ▼                ▼
 ┌──────────┐   ┌──────────┐
 │   L2     │   │   L4     │
 │  QDRANT  │   │ GRAPHITI │
 └──────────┘   └────┬─────┘
                     │
                     │ state transitions
                     ▼
               ┌──────────┐
               │   L5     │
               │HYSTERESIS│
               └────┬─────┘
                    │
                    │ ONLY SEAL-worthy events
                    ▼
               ┌──────────┐
               │   L6     │  SHA3-chained
               │VAULT999  │  F1-F13 gated
               │          │  append-only
               └──────────┘
              ◄──────── (no write-back path)
```

> ***Key Invariants:*** *L0 → L1 → L3 → L6 (data flows up, gets more durable). L2 and L4 are derived from L3 (can be rebuilt). L5 is meta-observation. L6 is cryptographically independent of L3. Once sealed, no path writes back downward.*

4. Tool Access Matrix

This is the **authorization policy** — which tools can read/write which layers.

TOOL	L0	L1	L2	L3	L4	L5	L6
Hermes (relay)	R/W	R/W	R	R	R	R	R
OpenClaw (infra/heavy)	R	R/W	R	R	R	R	R
APEX (judge)	R	R	R	R	R	R/W	R
OpenCode (executor)	R	R/W	—	R (project)	—	—	—
Curator (12h job)	—	R/W	R/W	R/W	R/W	R	—
Watcher (cron)	—	R/W	R	R/W	—	—	—
Arif (human)	R/W	R/W	R/W	R/W	R/W	R/W	R/W
`vault_seal`	R	—	—	R	—	—	**R/W**
`judge_deliberate`	R	R	R	R	R	R	R (read)

CRITICAL

OpenCode (the code execution tool, now in Telegram via `/forge`) has **NO access to L2–L6.** It works only in L0+L1 (intent + ephemeral) + project filesystem. This is by design — code tools should not have historical memory access, or they can fabricate provenance.

5. Recursive Improvement Loop

Every **12 hours**, the curator agent runs:

```
┌───────────────────────────────────────────────────────────┐
│  1. SCAN L1 for persistent items                          │
│     (conversations, sealed verdicts, working state)       │
│                                                           │
│  2. PROMOTE to L3 if:                                      │
│     - item has been touched > 5 times in L1                │
│     - or referenced in sealed decision (L6)                │
│     - or marked by agent as "important"                    │
│                                                           │
│  3. RE-EMBED L3 → L2                                       │
│     (update vectors for changed text)                      │
│                                                           │
│  4. SYNC L3 → L4                                           │
│     (update graph nodes/edges from row changes)            │
│                                                           │
│  5. COMPUTE L5 state                                       │
│     (compare previous state, detect transitions)           │
│                                                           │
│  6. REPORT deltas to Arif via Telegram                     │
│     (only if significant — silent = success)               │
└───────────────────────────────────────────────────────────┘
```

Watchers (RSS, GitHub, JSON APIs) run continuously and **deduplicate via watermark**. **Subagent depth-2 max** — leaf subagents cannot spawn their own children. **Darwinian evolver** refines underperforming prompts/code via evolutionary optimization.

6. Constitutional Floors F1–F13

The 13 floors wrap **every** tool call. They sit *between* L0 and L1, gating the transition from intent to action.

FLOOR	NAME	WHAT IT CHECKS
F1	AMANAH	Reversibility, integrity of action
F2	TRUTH	Claim evidence, no fabrication
F3	SEAL	Verdict finality
F4	SCOPE	Tool risk classification
F5	WITNESS	Audit trail present
F6	VOID	Rejection pathway
F7	HUMILITY	Uncertainty quantified
F8	SABAR	Patience for irreversible
F9	ANTI-HANTU	No deception, no pretense
F10	CLARITY	Context preservation
F11	DIGNITY	Human sovereignty preserved
F12	STEWARDSHIP	Resource constraint honored
F13	SOVEREIGN	Arif's signature + intent clear

T1 vs T4 bands:

- **T1 (autonomous):** no menu, no F13 hold required (commit audited diffs, run tests, edit configs)
- **T2–T3 (sabahan):** sabahan check
- **T4 (irreversible):** requires `888_HOLD` + Arif signature

7. Storage Stack & Cost

LAYER	TOOL	WHERE IT RUNS	COST
L0	Hermes session memory	hermes-asi-gateway (VPS)	$0
L1	Redis 7	VPS (Docker)	$5/mo
L2	Qdrant	VPS (Docker)	$0 (open source)
L3	Supabase	supabase.com (managed)	$25/mo (Pro tier)
L4	Graphiti (custom)	VPS (Docker)	$0
L5	Supabase + Redis LRU	supabase.com + VPS	$0
L6	JSONL files + Supabase mirror	VPS (append-only FS)	$0

Total infra cost: ~$30/mo (≈ RM140) for full federation.

> **PITCH MATERIAL**
>
> "Constitutional AI doesn't have to be expensive." Cost transparency is itself a design feature — no hidden token costs, no black-box inference bills, no vendor lock-in.

8. Why OpenCode Is Detached

The design choice to **isolate code-execution tools from persistent memory** is one of the most important architectural decisions in arifOS.

Risk of Memory-Connected Code Tools

If OpenCode could read/write L2–L6:

CAPABILITY	RISK
Read L3 transactions	Could "verify" its own lies against historical record
Read L6 seals	Could reference past decisions to justify present bad actions
Write L3 records	Could insert false evidence
Write L6 seals	Could fake constitutional verdicts
Read L4 graph	Could discover private reasoning paths
Read L5 hysteresis	Could "predict" system state and game it

The Detached Design

```
┌──────────────────────────────────┐
│  OpenCode authorized scope:      │
│                                  │
│  L0 · R (intent only)            │
│  L1 · R/W (working state)        │
│  Project filesystem · R/W        │
│  Network · sandboxed             │
│                                  │
│  No access to: L2, L3, L4, L5,   │
│                L6, secrets,      │
│                env vars outside  │
│                project scope     │
└──────────────────────────────────┘
```

What OpenCode can do:

- Read project files (source code, configs)
- Write project files (with audit log to L1)
- Run tests, linters, type checks
- Execute shell commands within project sandbox
- Spawn ephemeral subprocesses
- Send results back via Telegram

What OpenCode cannot do:

- Read VAULT999 seals
- Query historical decisions
- Reference past constitutional verdicts
- Insert records into Supabase
- Read Graphiti knowledge graph
- Modify memory in any persistent layer

> ***Arif's design philosophy:***
>
> "*A tool that knows history can lie about history. A tool that only knows the present cannot lie about the past — it can only be wrong about the present, and that's auditable.*"

The governance model is:

- **OpenCode = present-tense worker** (does the job, returns the result)
- **arifOS = historical witness** (records what happened, sealed to L6)
- **Arif = sovereign judge** (verifies, approves, seals)

If OpenCode proposes "I should commit X because last week we did Y," that's a **lie** — OpenCode has no way to know Y. The proposal is evaluated on its present merits only.

THE DIFFERENTIATOR

This is a **stronger** property than "AI agent with memory." It's a **constitutionally isolated** agent — fast, capable, but provably without historical fabrication power.

9. Real Examples

Example 1: Compose essay poster (this session)

```
Arif types in Telegram:
  /forge compose poster for "I hate AI, I hate DSG" essay
    ↓
L0: intent captured (envelope {intent, scope=opencode, risk=T1})
    ↓
L1: task queued, OpenCode session spawned
    ↓
OpenCode reads essay from project filesystem (read-only)
    ↓
OpenCode designs HTML, renders to PDF via weasyprint
    ↓
L1: working state — "poster generated, awaiting user approval"
    ↓
OpenCode returns PDF path to Hermes
    ↓
Hermes sends PDF to Telegram via MEDIA: protocol
    ↓
Arif: "bagus" → /seal
    ↓
L6: sealed (T1, reversible but logged for audit)
```

Notice: OpenCode never touched L2–L6. The seal happened at the Hermes level, with constitutional gating. OpenCode is a worker; Hermes is the witness.

Example 2: WEALTH NPV analysis (future capability)

```
Arif: /ask "what's NPV for Q3 portfolio given current rates?"
    ↓
L0: intent captured
L1: routing decision — query routes to WEALTH organ (port 18082)
    ↓
WEALTH MCP receives request:
  - reads L3 portfolio data (current positions)
  - reads L3 macro rates (from L8 field)
  - calls wealth_synthesize (Ω-00)
    ↓
WEALTH returns: {p10: -2.3M, p50: 8.1M, p90: 24.7M, verdict: SABAR}
    ↓
arifOS receives verdict:
  - checks F2 (truth: claims backed by data)
  - checks F7 (humility: uncertainty bands present)
  - checks F1 (reversibility: this is a calculation, not a commitment)
    ↓
Hermes returns: "SABAR. P50 = RM8.1M, downside RM2.3M. Want me to seal this analysis?"
    ↓
Arif: "ya"
    ↓
L6: sealed (SABAR verdict, audit trail complete)
```

10. Maintenance & Operations

CADENCE	ACTIVITY
Daily	Curator agent runs at 00:00 MYT (12h after last run); watchers poll configured feeds; L1 → L3 promotion
Weekly	L2 vector re-embed (incremental); L4 graph consistency check; L5 hysteresis state evaluation
Monthly	L6 chain verification (full walk); backup of all layers to cold storage; capacity planning review
Quarterly	Architectural review; constitutional layer audit (F1–F13 still relevant?); tool access matrix review

11. Roadmap 2026–2027

2026 Q3:

- L6 cryptographic signature via Arif's hardware key (YubiKey)
- L4 graph RAG (combine vector + graph for better context)
- L5 hysteresis auto-actions (e.g. auto-degrade quality when in REVERSION)

2026 Q4:

- Multi-federation isolation (separate VAULT999 chains per project)
- On-chain anchoring of L6 (publish SHA3 root daily to public ledger)
- L1 distributed cache (Redis Cluster)

2027:

- Self-evolving constitutional floors (F1–F13 can update via sealed verdict)
- L5 ML-based state prediction (not just rule-based)
- Cross-tenant memory isolation for SaaS deployment

12. Glossary

TERM	DEFINITION
Envelope	Structured intent wrapper: {actor, intent, scope, reversibility, risk_band, payload, timestamp}
Seal	Cryptographically-signed, constitutionally-gated entry in VAULT999
Witness	Agent or human who attests to a decision
Floor	Constitutional check (F1–F13) that gates tool calls
888_HOLD	Pause-and-confirm gate for irreversible or high-impact actions
SABAR	Verdict meaning "wait, more evidence needed"
VOID	Verdict meaning "rejected, evidence insufficient"
SEAL	Verdict meaning "approved, irreversible, recorded"
Hysteresis	Path-dependent state (system remembers where it came from)
Petala	Layer (from Malay/Jawi — "petala langit" = layer of the sky)
DSG	Digital Sovereign Governance — every action passes through independent authority-bearing runtime
MCP	Model Context Protocol — standard interface for AI agents to call external tools

DITEMPA BUKAN DIBERI · 999 SEAL ALIVE

arifOS Memory Architecture Reference · v2.0 · 2026-06-07
Authored by: Muhammad Arif bin Fazil · for Fahmi (technical onboarding)
Constitutional Floor: F1 AMANAH · F2 TRUTH · F9 ANTI-HANTU · all passed